

2	(a) (i) $v^2 = u^2 + 2as$ $= (8.4)^2 + 2 \times 9.81 \times 5$ $= 12.99 \text{ m s}^{-1}$ (allow 13 to 2 s.f. but not 12.9)	C1	A1	[2]
	(ii) $t = (v - u) / a$ or $s = ut + \frac{1}{2}at^2$ $= (12.99 - 8.4) / 9.81$ or $5 = 8.4t + \frac{1}{2} \times 9.81t^2$ $t = 0.468 \text{ s}$	M1	A0	[1]
	(b) reasonable shape suitable scale correctly plotted 1 st and last points at (0,8.4) and (0.88 – 0.96,0) with non-vertical line at 0.47 s	M1	A1	[3]
	(c) (i) 1. kinetic energy at end is zero so $\Delta KE = \frac{1}{2}mv^2$ or $\Delta KE = \frac{1}{2}mu^2 - \frac{1}{2}mv^2$ $= \frac{1}{2} \times 0.05 \times (8.4)^2$ $= (-) 1.8 \text{ J}$	C1	A1	[2]
	2. final maximum height = $(4.2)^2 / (2 \times 9.8) = (0.9 \text{ (m)})$ change in PE = $mgh_2 - mgh_1$ $= 0.05 \times 9.8 \times (0.9 - 5)$ $= (-) 2.0 \text{ J}$	C1	C1	[3]
	(ii) change is – 3.8 (J) energy lost to ground (on impact) / energy of deformation of the ball / thermal energy in ball	B1	B1	[2]

3 (a) A body continues at rest or constant velocity unless acted on by a resultant